BRAC UNIVERSITY
Department of Computer Science and Engineering

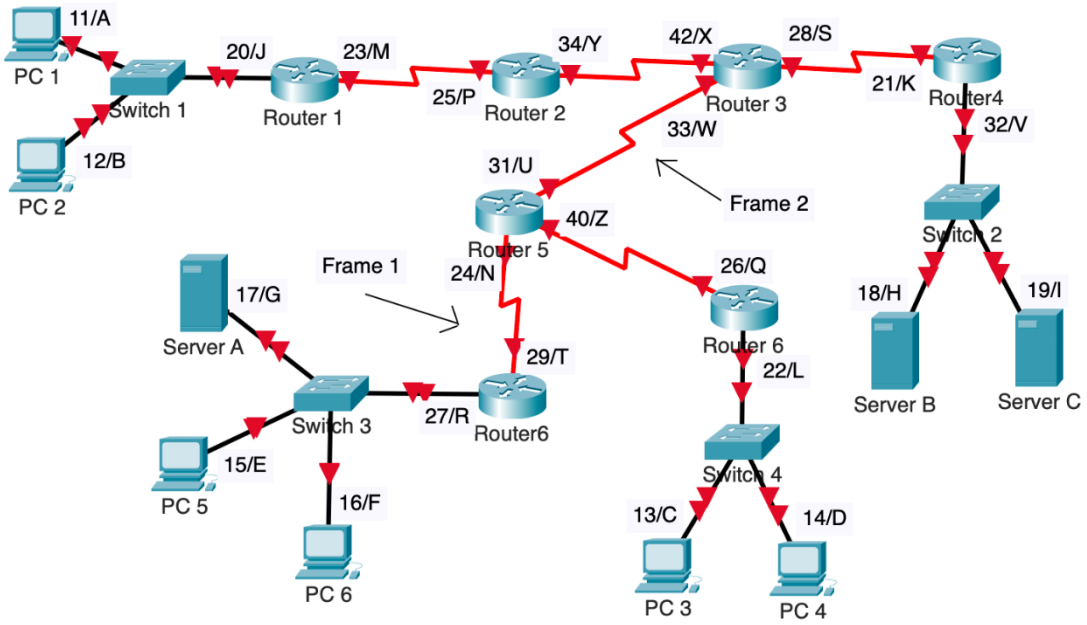
Examination: Semester Midterm | Semester: Summer 2024
Duration: 1 hour 20 min | Full Marks: 45

CSE 320: Data Communications

Answer the following questions.
Figures in the right margin indicate marks.

SET B

Name:	ID:	Section:

1. [CO1]	a)	Suppose a university campus has four departments. Each department has a faculty area and a student area. Each area can communicate with each other as they are connected centrally using a switch. If one area gets disconnected, other areas remain active. Again, the devices in both areas are connected using multipoint connections only. Finally, each department has connections with only the neighboring departments. **Identify** the topologies from the above scenario and **draw** the hybrid topology, assuming there are 3 nodes in each area. **Count** the total number of communication links in this hybrid topology.	[3+2]
	b)	Consider the network topology shown below. Assume the physical addresses to be the uppercase letters and the logical addresses to be the numbers. For the Devices, use a Port number from the dynamic range (49152 - 65535).  i. How many networks are there in total?	[1+5]

ii. Data is coming from Server B which uses port number 23 and it is intended for PC6. Considering this situation, **complete** Frame 1 below by writing the appropriate Destination and Source Physical, Logical and Port Addresses.

iii. PC1 is trying to send a data to Server A, which is an Email Server using port 25. Considering this situation, **complete** Frame 2 below by writing the appropriate Destination and Source Physical, Logical and Port addresses.

	D. Mac	S. Mac	D. IP	S. IP	D. Port	S. Port		
Frame 1							Data	Trailer

	D. Mac	S. Mac	D. IP	S. IP	D. Port	S. Port		
Frame 2							Data	Trailer

c) Annabeth is **composing an email [1]** to send to her friend. This application **ensures a session [2]** with the email server and the message will be transmitted securely. Once prepared, the message is segmented into smaller packets and **properly routed [3]** across networks to reach its destination. At the same time, the app manages the **physical connection to the network interface [4]**, ensuring seamless transmission of data bits over the network medium.

Identify the layers of TCP/IP model which are responsible for the phrases in bold. [4]

2. [CO2]

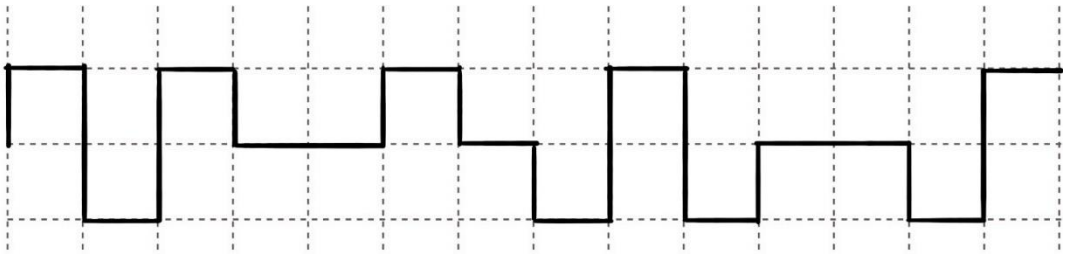
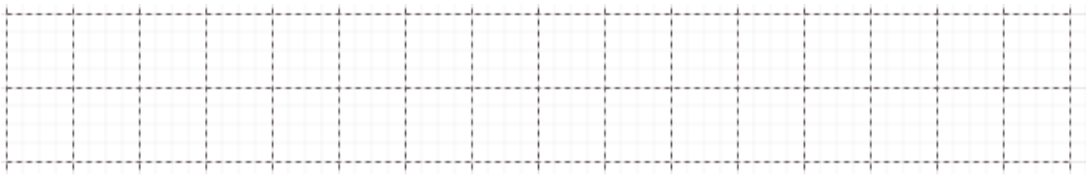
a) **What** is the difference between baseband and broadband transmission? **Which** type of channel is used by these transmission techniques?
If a periodic signal having lowest frequency of 150 kHz has a bandwidth of 45000 Hz, then calculate the highest frequency contained in that signal. **Draw** the frequency domain diagram of this signal whose maximum amplitude is 15 V. [3+3]

b) Suppose you are sending a doc file containing 9 pages at a speed of 48kbps through a noiseless channel. Each page includes 56 characters and the doc file has been encoded using ASCII [8 bit]. The channel through which the file is being shared has 14 levels. The signal strength of the channel is 135 W. [2+3+2]

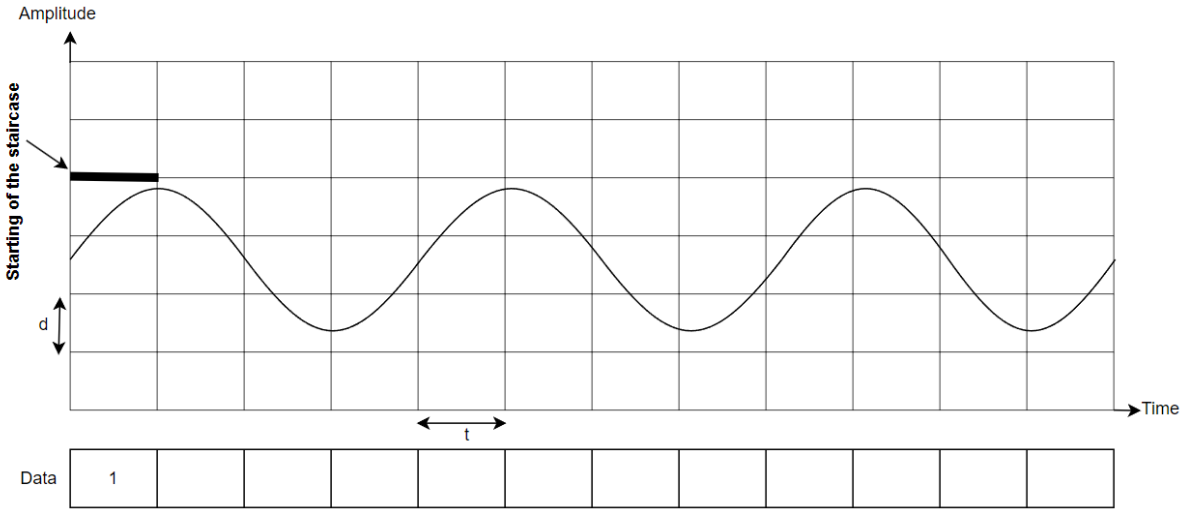
i. **Calculate** the bandwidth of the channel in Hz.

ii. If the capacity of the channel is 20% less than the noiseless channel's bitrate, **calculate** the power of noise.

iii. **Find** out the transmission delay to send the file if the noiseless channel's bitrate is considered as the bandwidth of the device.

	c)	**Define** noise. If the value of SNR is infinite, what will be the value of signal power and the noise power?	[2]
3. [CO2]	a)	To minimize the consecutive zero problem, we are using the following block coding scheme shown in the table. Based on the scheme, answer the questions below: • What will be the original bitstream for the following encoded bit stream: 0111101010100101 • Calculate the percentage of redundant bits required for the scheme. • For any encoded sequence, what will be the maximum number of consecutive 0's if we use this block coding scheme?	[5]
3. [CO2]	b)	The signal is drawn using HDB3 scrambling technique. • **Decode** the signal and write the bit stream.  • Now **draw** a digital signal using the Differential Manchester line coding scheme for the decoded bit stream. (sketch the signal in the question paper only) 	[5]
	c)	Does the bit rate change for changing the number of levels in the PCM technique? Yes or No. Write briefly.	[2]

Data Sequence	*Encoded Sequence*
000	1010
001	1011
010	0111
011	1101
100	1110
101	1111
110	0101
111	1100

	d)	**Draw** the staircase diagram and **generate the binary data** simultaneously in the below analog-to-digital modulation technique. (sketch it in the question paper only)  	[3]